

Some Properties of Cyclotomic Polynomials

Mobius Inversion Formula

Define the *Mobius Function* $\mu: \mathbb{Z}^+ \rightarrow \{0, 1, -1\}$ as follows:

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n \text{ is square free and } k \text{ is the number of prime divisor of } n. \\ 0 & \text{otherwise} \end{cases}$$

Clearly μ is *multiplicative*, namely if m and n are relatively prime, then $\mu(mn) = \mu(m)\mu(n)$. Moreover

Theorem 1: Let n be a positive integer. Then $\sum_{d|n} \mu(d) = 1$ for $n = 1$, and

$$\sum_{d|n} \mu(d) = 0 \text{ for } n \geq 2.$$

Proof. For $n = 1$, it is clear. Now define $T = \prod_{\substack{p|n \\ p \text{ prime}}} p$ as the product of all distinct prime divisors of n . Then it is clear $\sum_{d|n} \mu(d) = \sum_{d|T} \mu(d)$. But suppose now T has k distinct prime divisors, i.e., $T = p_1 p_2 \dots p_k$, then

$$\sum_{d|T} \mu(d) = \sum_{l=0}^k \binom{k}{l} (-1)^l = (1-1)^k = 0.$$

Now we have

Theorem 2 (Mobius Inversion Formula): Suppose $F, f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ are functions so that $F(n) = \sum_{d|n} f(d)$. Then $f(n) = \sum_{d|n} \mu(d) F(\frac{n}{d})$.

Proof. Note $\sum_{d|n} \mu(d) F(\frac{n}{d}) = \sum_{d|n} \mu(d) \sum_{\substack{t|n \\ t|\frac{n}{d}}} f(t)$. Now t is a divisor of $\frac{n}{d}$ and is a

divisor of n . Thus

$$\sum_{d|n} \mu(d) \sum_{\substack{t|\frac{n}{d}}} f(t) = \sum_{t|n} f(t) \sum_{\substack{d|n \\ t|\frac{n}{d}}} \mu(d) = \sum_{t|n} f(t) \sum_{\substack{d|\frac{n}{t}}} \mu(d). \text{ But the term } \sum_{\substack{d|\frac{n}{t}}} \mu(d) \text{ is 1 when}$$

$n = t$ and 0 otherwise. Hence we are done.

Remark: The multiplicative version of the theorem is as follows. Given f ,

$F: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ such that $F(n) = \prod_{d|n} f(d)$, then $f(n) = \prod_{d|n} F(\frac{n}{d})^{\mu(d)}$.

$$\text{Proof. } \prod_{d|n} F(\frac{n}{d})^{\mu(d)} = \prod_{d|n} \left(\prod_{t|\frac{n}{d}} f(t) \right)^{\mu(d)} = \prod_{t|n} f(t)^{\sum_{d|\frac{n}{t}} \mu(d)} = \prod_{t|n} f(t)^{\sum_{r|\frac{n}{t}} \mu(r)} = f(n).$$

Primitive Roots of Unity

A complex number ξ is an n th root of unity if $\xi^n = 1$. We know they are $e^{\frac{2\pi i}{n}}, e^{\frac{4\pi i}{n}}, \dots, e^{\frac{2n\pi i}{n}}$, where $e^{\frac{2k\pi i}{n}} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$. If ξ is an n th root of unity, then the smallest positive integer k such that $\xi^k = 1$ is the *order* of ξ , denoted by $\text{ord}(\xi)$. Easy to show the **Facts**:

1. If ξ is an n th root of unity and $\xi^k = 1$, using division algorithm, $\text{ord}(\xi) | k$. In

particular $\text{ord}(\xi) | n$.

2. Let ξ be an n th root of unity, then $\xi^k = \xi^l$ if and only if $k \equiv l \pmod{\text{ord}(\xi)}$. This simply follows from $\xi^{k-l} = 1$.

Now we define: if ξ is an n th root of unity. Then ξ is a **primitive n th root of unity** if $\text{ord}(\xi) = n$. Indeed if $\text{ord}(\xi) = d$, then ξ is a primitive d root of unity. We now observe other **Facts**:

3. If ξ is a primitive n th root of unity, then $\xi, \xi^2, \dots, \xi^n = 1$ are all n th (distinct) roots of unity. They are all distinct roots of $x^n = 1$.

4. If ξ is a primitive n th root of unity, then ξ^k is a primitive n th root of unity if and only if $\gcd(n, k) = 1$. Hence there are $\phi(n)$ (Euler's totient function) of them.

Indeed suppose $\text{ord}(\xi^k) = d$, then $\xi^{kd} = 1$, hence $n | kd$. So if $\gcd(n, k) = 1$, then

$n | d$, thus $n = d$. On the other hand, $\gcd(n, k) \neq 1$, then $\xi^{\frac{k}{\gcd(n, k)} \frac{n}{\gcd(n, k)}} = 1$, so $d < n$, or ξ^k not primitive.

5. Again assume ξ is a primitive n th root of unity, t a factor of n , then ξ^t is of order $\frac{n}{t}$, and there are $\phi(\frac{n}{t})$ elements of such order. This gives a proof of

$$\sum_{d|n} \phi(d) = n.$$

Example 1: From $\sum_{d|n} \phi(d) = n$, and the Inversion formula, we have

$$\phi(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) d = \sum_{t|n} \mu(t) \frac{n}{t} = n \sum_{t|n} \frac{\mu(t)}{t}.$$
 Suppose now $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$, then

$$\begin{aligned} \sum_{t|n} \frac{\mu(t)}{t} &= 1 - \sum_i \frac{1}{p_i} + \sum_{i < j} \frac{1}{p_i p_j} - \dots + (-1)^r \frac{1}{p_1 p_2 \dots p_r} \\ &= \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_r}\right). \end{aligned}$$

We then have a formula of the Euler's totient function.

Cyclotomic Polynomials

The ***n*th cyclotomic polynomial**, $\Phi_n(x)$, is the (monic) polynomial having exactly the primitive *n*th roots of unity as roots, that is,

$$\Phi_n(x) = \prod_{\substack{\xi^n = 1 \\ \text{ord}(\xi) = n}} (x - \xi).$$

As discussed, we see

Theorem 3: $x^n - 1 = \prod_{d|n} \Phi_d(x).$

Check that the two sides have the same roots and both are monic. Comparing the x^n terms, we also have $n = \sum_{d|n} \phi(d)$. (The degree of $\Phi_d(x)$ is $\phi(d)$).

Using the Inversion Formula, we have

Theorem 4: $\Phi_n(x) = \prod_{d|n} (x^{\frac{n}{d}} - 1)^{\mu(d)}.$

Example 2: Using Theorem 3 and 4, we get

$$\Phi_1(x) = x - 1$$

$$\Phi_2(x) = \frac{x^2 - 1}{x - 1} = x + 1$$

$$\Phi_3(x) = \frac{x^3 - 1}{x - 1} = x^2 + x + 1$$

$$\Phi_4(x) = \frac{(x^4 - 1)}{x^2 - 1} = x^2 + 1$$

$$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$$

$$\Phi_6(x) = x^2 - x + 1$$

$$\Phi_7(x) = x^6 + x^5 + \dots + 1$$

$$\Phi_8(x) = x^4 + 1$$

$$\Phi_9(x) = x^6 + x^3 + 1$$

$$\Phi_{10}(x) = x^4 - x^3 + x^2 - x + 1$$

$$\Phi_{11}(x) = x^{10} + x^9 + \dots + x + 1$$

$$\Phi_{12}(x) = x^4 - x^2 + 1$$

$$\Phi_{13}(x) = x^{12} + x^{10} + \dots + x + 1$$

$$\Phi_{14}(x) = x^6 - x^5 + x^4 - x^3 + x^2 - x + 1$$

$$\Phi_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$$

$$\Phi_{16}(x) = \frac{x^{16} - 1}{x^8 - 1} = x^8 + 1$$

$$\Phi_{17}(x) = x^{16} + x^{15} + \dots + x + 1$$

$$\Phi_{18}(x) = \Phi_9(-x) = x^6 - x^3 + 1 \quad (\text{see subsequent theorems})$$

$$\Phi_{19}(x) = x^{18} + x^{17} + \dots + x + 1$$

$$\Phi_{20}(x) = \frac{(x^{20} - 1)(x - 1)(x + 1)}{(x^4 - 1)(x^{10} - 1)} = x^8 - x^7 + x^6 - x^5 + x^4 - x^3 + x^2 - x + 1$$

Using Theorem 3, 4 and some computer software, we also calculate some other

values:

$$\begin{aligned}\Phi_{21}(x) &= \frac{(x^{21}-1)(x-1)}{(x^3-1)(x^7-1)} = \frac{x^{14}+x^7+1}{x^2+x+1} \\ &= x^{12}-x^{11}+x^9-x^8+x^6-x^4+x^3-x+1.\end{aligned}$$

$$\begin{aligned}\Phi_{22}(x) &= \frac{(x^{22}-1)(x-1)}{(x^2-1)(x^{11}-1)} = \frac{x^{11}+1}{x+1} \\ &= x^{10}-x^9+x^8-x^7+x^6-x^5+x^4-x^3+x^2-x+1.\end{aligned}$$

$$\begin{aligned}\Phi_{24}(x) &= \frac{(x^{24}-1)}{(x^{12}-1)(x^4+1)} = \frac{x^{12}+1}{x^4+1} \\ &= x^8-x^4+1.\end{aligned}$$

$$\begin{aligned}\Phi_{26}(x) &= \frac{(x^{26}-1)}{(x^{13}-1)(x+1)} = \frac{x^{13}+1}{x+1} \\ &= x^{12}-x^{11}+x^{10}-x^9+x^8-x^7+x^6-x^5+x^4-x^3+x^2-x+1.\end{aligned}$$

$$\begin{aligned}\Phi_{28}(x) &= \frac{(x^{28}-1)(x-1)(x+1)}{(x^4-1)(x^{14}-1)} \\ &= x^{12}-x^{10}+x^8-x^6+x^4-x^2+1.\end{aligned}$$

$$\begin{aligned}\Phi_{30}(x) &= \frac{(x^{30}-1)(x-1)(x^2+x+1)}{(x^6-1)(x^{15}-1)(x^4-x^3+x^2-x+1)} \\ &= x^8-x^7-x^5-x^4-x^3+x+1.\end{aligned}$$

$$\begin{aligned}\Phi_{33}(x) &= \frac{(x^{33}-1)(x-1)}{(x^3-1)(x^{11}-1)} = \frac{x^{22}-x^{11}+1}{x^2-x+1} \\ &= x^{20}+x^{19}-x^{17}-x^{16}+x^{14}+x^{13}-x^{11}-x^{10} \\ &\quad -x^9+x^7+x^6-x^4-x^3+x+1.\end{aligned}$$

$$\begin{aligned}\Phi_{40}(x) &= \frac{x^{40}-1}{(x^{20}-1)(x^4+1)} = \frac{x^{20}+1}{x^4+1} \\ &= x^{16}-x^{12}+x^8-x^4+1.\end{aligned}$$

In cases of powers of 2, we have

$$\Phi_2(x) = x+1.$$

Hence

$$\Phi_4(x) = \frac{x^4 - 1}{x^2 - 1} = x^2 + 1, \Phi_8(x) = \frac{x^8 - 1}{x^4 - 1} = x^4 + 1, \Phi_{16}(x) = \frac{x^{16} - 1}{x^8 - 1} = x^8 + 1,$$

$$\Phi_{32}(x) = \frac{x^{32} - 1}{x^{16} - 1} = x^{16} + 1,$$

etc.

Also

$$\Phi_3(x) = \frac{x^3 - 1}{x - 1} = x^2 + x + 1.$$

Hence

$$\Phi_9(x) = \frac{x^9 - 1}{x^3 - 1} = x^6 + x^3 + 1.$$

And

$$\Phi_{27}(x) = \frac{x^{27} - 1}{x^9 - 1} = x^{18} + x^9 + 1, \text{ etc.}$$

Finally

$$\begin{aligned} \Phi_{105}(x) &= \frac{(x^{105} - 1)(x^3 - 1)(x^5 - 1)(x^7 - 1)}{(x^{15} - 1)(x^{21} - 1)(x^{35} - 1)(x - 1)} \\ &= x^{48} + x^{47} + x^{46} - x^{43} - x^{42} - 2x^{41} - x^{40} - x^{39} + x^{36} + x^{35} + x^{34} + x^{33} \\ &\quad + x^{32} + x^{31} - x^{28} - x^{26} - x^{24} - x^{22} - x^{20} + x^{17} + x^{16} + x^{15} + x^{14} + x^{13} \\ &\quad + x^{12} - x^9 - x^8 - 2x^7 - x^6 - x^5 + x^2 + x + 1. \end{aligned}$$

This is the first n that coefficients other than 0, 1 or -1 appears.

Example 3:

1. For an odd prime p , $\Phi_p(x) = x^{p-1} + x^{p-2} + \dots + 1$.

2. For prime power p^r , $\Phi_{p^r}(x) = \frac{x^{p^r} - 1}{x^{p^{r-1}} - 1} = x^{p^{r-1}(p-1)} + x^{p^{r-1}(p-2)} + \dots + x^{p^{r-1}} + 1$.

3. If n is odd larger than 1, then $\Phi_{2n}(x) = \Phi_n(-x)$. In particular for an odd prime p ,

$$\Phi_{2p}(x) = x^{p-1} - x^{p-2} + \dots - x + 1.$$

Proof: (1). Change x to $x+1$ and apply the Eisenstein's criteria. (2) Compare

$x^{p^r} - 1$ and $x^{p^{r-1}} - 1$. (3) Easy to check, if $\omega_1, \omega_2, \dots, \omega_s$ are the primitive n^{th}

roots of unity, then $-\omega_1, -\omega_2, \dots, -\omega_s$ are the primitive $2n^{\text{th}}$ roots of unity. Hence

$$\Phi_n(-x) = \prod_{k=1}^s (-x - \omega_k) = (-1)^s \prod_{k=1}^s (x + \omega_k) = \Phi_{2n}(x). \text{ Note } s \text{ is even.}$$

We give some properties of cyclotomic polynomials. Before all that, a lemma:

Lemma 1: Suppose $f(x) = x^m + a_{m-1}x^{m-1} + \dots + a_0$ and

$g(x) = x^n + b_{n-1}x^{n-1} + \dots + b_0$ are polynomials with rational coefficients. If all

coefficients of $f(x)g(x)$ are integers, then so are $f(x)$ and $g(x)$.

Proof. Let M and N be the smallest positive integers so that all coefficients of $Mf(x)$ and $Ng(x)$ are integers. Let $A_i = Ma_i, 0 \leq i \leq m-1$ and $A_m = M$; also

$B_j = Nb_j, 0 \leq j \leq n-1, B_n = N$. Then $MNf(x)g(x) = A_m B_n x^{m+n} + \dots + A_0 B_0$ will

consist only integer coefficients. Suppose $MN > 1$ and p a prime divisor of MN .

(As $f(x)g(x)$ is a polynomial with integer coefficients, then $MNf(x)g(x)$ is a polynomial with integer coefficients and every coefficient is divisible by p). We claim there exists an $i, 0 \leq i \leq m$, such that $p \nmid A_i$. Indeed if $p \nmid M$, then

$p \nmid A_m$ and if $p \mid M$, then $p \nmid A_i$ (?) for all $i, 0 \leq i \leq m$, then $\frac{A_i}{p} = \frac{M}{p} a_i$ an

integer, contradicting the minimality of M . Likewise there exists an integer j such

that $p \nmid B_j$. Let I and J be the greatest of these i and j . Then the coefficient of

x^{I+J} in $MNf(x)g(x)$ is $A_I B_J + pR$ not divisible by p , contradiction again, as

coefficients of $MNf(x)g(x)$ should be divisible by p . Thus $MN = 1$.

(Remark: This is the Gauss' Lemma).

Theorem 5:

1. The coefficients of $\Phi_n(x)$ are integers.
2. $\Phi_n(0) = 1, n \geq 2$.
3. $\Phi_n(x) > 0$ for all real x .
4. $\Phi_n(x)$ is strictly increasing for $x > 1$.
5. For all $x > 0$, $(x-1)^{\phi(n)} < \Phi_n(x) < (x+1)^{\phi(n)}$.

6. Let $n \geq 2$. Let $\Phi_n(x) = \sum_{i=0}^m c_i x^i$. Then (i) $\Phi_n(x) = x^m \Phi_n(1/x)$ for $x \neq 0$, (ii) $c_{m-r} = c_r$ for $0 \leq r \leq m$. (Here of course $m = \phi(n)$.)
7. $\Phi_n(x)$ are irreducible over $\mathbb{Z}$.

Proof.

1. By induction $\Phi_1(x) = x - 1$. Suppose that the statement is true for all $k < n$. Then

$$\Phi_n(x) = \frac{x^n - 1}{\prod_{\substack{d|n, \\ d \neq n}} \Phi_d(x)}.$$

The coefficients of $\Phi_n(x)$ are rational (division algorithm)

and thus are integers by Lemma 1.

2. We note the roots of $x^n - 1$ are 1, -1 (when n even) and complex roots appear in

$$\text{pairs } e^{\frac{2k\pi i}{n}} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} \quad \text{and} \quad e^{\frac{2(n-k)\pi i}{n}} = \cos \frac{2(n-k)\pi}{n} + i \sin \frac{2(n-k)\pi}{n}.$$

(Note they are of the same order),

$$\text{hence } (x - e^{\frac{2k\pi i}{n}})(x - e^{\frac{2(n-k)\pi i}{n}}) = x^2 - 2x \cos \frac{2k\pi}{n} + 1. \quad \text{So } \Phi_n(0) = 1. \quad (\text{Note } \pm 1 \text{ will}$$

not appear.)

(Remark: using this, can show the cyclotomic polynomials are “symmetric”.)

3. 4. For $-1 < a < 1$, write $f_a(x) = x^2 - 2ax + 1$. Then

$$x^2 - 2ax + 1 = (x - a)^2 + (1 - a^2) > 0 \quad \text{and is strictly increasing for all } x \geq 1. \quad \text{Thus (3) and (4).}$$

5. Also $(x - 1)^2 < x^2 - 2ax + 1 < (x + 1)^2$ for all $x > 0$. Thus (5).

6. The statement is true for $n = 2$. For $n > 2$, write $a_k = \cos \frac{2k\pi}{n}$, and by putting

in suitable pairs, we have

$$x^m \Phi_n\left(\frac{1}{x}\right) = \prod x^2 \left(1 - \frac{2a_k}{x} + \frac{1}{x^2}\right) = \prod (x^2 - 2a_k x + 1) = \Phi_n(x). \quad \text{Hence}$$

$$\sum_{i=0}^m c_i x^i = \sum_{i=0}^m c_i x^{m-i} = \sum_{j=0}^m c_{m-j} x^j = \sum_{i=0}^m c_{m-i} x^i \quad \text{for all } x \neq 0. \quad \text{Thus } c_r = c_{m-r}$$

(palindrome).

7. See Jacobson or Jameson. For $n = p$, may use Eisenstein. Thus we say $\Phi_n(x)$ is the *minimal polynomial* of a primitive n th root of unity ξ . We can proceed to prove the statement roughly as follows.

First the *claim*: Let $g(x)$ be an irreducible divisor of $\Phi_n(x)$ in $\mathbb{Z}[x]$ and let ζ be a complex root of $g(x)$. Then for each prime p with $\gcd(n, p) = 1$, then the complex number ζ^p is also a root of $g(x)$.

Let the claim be confirmed, say. Let us take $\zeta_0 = e^{\frac{2\pi i}{n}}$, it is clearly a primitive n^{th} root of 1, so ζ_0 is a root of $\Phi_n(x)$, hence a root of some irreducible divisor $g(x)$ of $\Phi_n(x)$ in $\mathbb{Z}[x]$. From the statement above, for any prime p_1 not dividing n , the complex number $\zeta_1 = \zeta_0^{p_1}$ is also a root of $g(x)$. By the same token, for any prime p_2 not dividing n , the complex number $\zeta_2 = \zeta_1^{p_2} = \zeta_0^{p_1 p_2}$ is also a root of $g(x)$, etc., so for any collection of (not necessarily distinct) primes $p_1, p_2, \dots, p_k$ not dividing n , the number $\zeta_0^{p_1 p_2 \dots p_k}$ is also a root of $g(x)$. But all primitive roots of 1 are of the form ζ_0^k with $\gcd(k, n) = 1$, so all primitive n^{th} root of 1 are roots of $g(x)$ and $g(x) = \Phi_n(x)$.

Thus it remains to show the validity of the claim. We proceed by contradiction. Let $\Phi_n(x) = g(x)h(x)$. As before ζ is a root of $g(x)$ but ζ^p is a root of $h(x)$. Clearly ζ^p is also a primitive n^{th} root of 1 and note that ζ is a root of $h(x^p)$. This means $g(x)$ and $h(x^p)$ have a common root. But as $g(x)$ is irreducible, $h(x^p)$ is divisible by $g(x)$. Now we consider all polynomials modulo p . We note that using the Fermat's little theorem, $h(x^p) = (h(x))^p \pmod{p}$. This implies there exists an irreducible factor $g_1(x)$ of $g(x)$ and $h(x) \pmod{p}$. Now $\Phi_n(x) = g(x)h(x) \pmod{p}$, hence $\Phi_n(x)$ is divisible by $g_1^2(x) \pmod{p}$. This implies $x^n - 1$ is divisible by $g_1^2(x) \pmod{p}$. But this is a contradiction since the derivative of $x^n - 1$ is nx^{n-1} which is not divisible by p .

Theorem 6.

1. Let p be a prime number and n a positive integer. Then

$$\Phi_{pn}(x) = \begin{cases} \Phi_n(x^p) & \text{if } p \mid n \\ \frac{\Phi_n(x^p)}{\Phi_n(x)} & \text{if } p \nmid n. \end{cases}$$

2. Let p be a prime number, n and k are positive integer. Then

$$\Phi_{p^k n}(x) = \begin{cases} \Phi_n(x^{p^k}) & \text{if } p|n \\ \frac{\Phi_n(x^{p^k})}{\Phi_n(x^{p^{k-1}})} & \text{if } p \nmid n. \end{cases}$$

Proof. 1. Suppose $p|n$. Then

$$\begin{aligned} \Phi_{pn}(x) &= \prod_{d|pn} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \\ &= \left(\prod_{d|n} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \left(\prod_{\substack{d|pn \\ d \nmid n}} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \\ &= \Phi_n(x^p) \prod_{\substack{d|pn \\ d \nmid n}} (x^{\frac{pn}{d}} - 1)^{\mu(d)}. \end{aligned}$$

As $d|pn, d \nmid n$ and $p|n$, implies that $p^2|d$, so $\mu(d) = 0$. Hence the result.

Suppose now $p \nmid n$. Then

$$\begin{aligned} \Phi_{pn}(x) &= \prod_{d|pn} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \\ &= \left(\prod_{d|n} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \left(\prod_{d|pn} (x^{\frac{pn}{pd}} - 1)^{\mu(d)} \right) \\ &= \left(\prod_{d|n} (x^{\frac{pn}{d}} - 1)^{\mu(d)} \right) \left(\prod_{d|n} (x^{\frac{n}{d}} - 1)^{-\mu(d)} \right) \\ &= \frac{\Phi_n(x^p)}{\Phi_n(x)}. \end{aligned}$$

2. Hence

$$\Phi_{p^k n}(x) = \Phi_{p^{k-1}n}(x^p) = \dots = \Phi_{pn}(x^{p^{k-1}}) = \begin{cases} \Phi_n(x^{p^k}) & \text{if } p|n \\ \frac{\Phi_n(x^{p^k})}{\Phi_n(x^{p^{k-1}})} & \text{if } p \nmid n. \end{cases}$$

Lemma 2:

1. Let p be a prime number. Suppose $x^n - 1$ has a double root modulo p , i.e., $x^n - 1 \equiv (x-a)^2 f(x) \pmod{p}$. Then $p|n$. (Here a is an integer and $f(x)$ is a polynomial with integer coefficients.)
2. Let $d|n$ and $d < n$, x be an integer. Suppose p is a common divisor of $\Phi_d(x)$

and $\Phi_n(x)$. Then $p|n$.

Proof. 1. Clearly $p \nmid a$, otherwise $-1 \equiv 0 \pmod{p}$. Taking derivatives, get

$$nx^{n-1} \equiv 2(x-a)f(x) + (x-a)^2 f'(x) \pmod{p}, \text{ or } na^{n-1} \equiv 0 \pmod{p}. \text{ Hence } p|n.$$

2. $x^n - 1 = \prod_{i|n} \Phi_i(x)$, and p divides $\Phi_d(x)\Phi_n(x)$, thus $x^n - 1$ has a double root

and the result.

Theorem 7

1. Let x be an integer. Then every prime factor p of $\Phi_n(x)$ either satisfies

$$p \equiv 1 \pmod{n} \text{ or } p|n.$$

2. Let p be a prime number. Then every prime factor q of $1 + x + \dots + x^{p-1}$ either satisfies $q \equiv 1 \pmod{p}$ or $q = p$.

Proof.

1. Let p be a prime divisor of $\Phi_n(x)$. Note $p \nmid x$ since $p|\Phi_n(x)|x^n - 1$. Let

$$k = \text{ord}_n(x). \text{ Since } p|x^n - 1, \text{ we have } x^n \equiv 1 \pmod{p}, \text{ so } k|n. \text{ If } k = n, \text{ then}$$

$$n|p-1, \text{ i.e., } p \equiv 1 \pmod{n}, \text{ since } x^{p-1} \equiv 1 \pmod{p} \text{ by Fermat's Little Theorem.}$$

Suppose now $k < n$. Since $0 \equiv x^k - 1 = \prod_{d|k} \Phi_d(x) \pmod{p}$, there exists a divisor d

of k such that $p|\Phi_d(x)$. But $d|k|n$ and $d < n$, so from Lemma 2.2, the result follows.

2. Note $\Phi_p(x) = 1 + x + x^2 + \dots + x^{p-1}$.

Example 4 Check the prime factors of $1 + x + \dots + x^4 = \Phi_5(x)$ are divisible by 5 or

$$\equiv 1 \pmod{5}. \text{ Also } \Phi_8(x) = x^4 + 1. \text{ The prime factors are either divisible by 2 or}$$

$$\equiv 1 \pmod{8}. \text{ Check } 8^4 + 1 = 17 \times 241. \text{ Now } \Phi_{72}(x) = x^{24} - x^{12} + 1. \text{ Check}$$

$$2^{24} - 2^{12} + 1 = 433 \times 38737 \text{ and each prime factor } \equiv 1 \pmod{72}.$$

Other examples: $\Phi_5(4) = 4^4 + 4^3 + 4^2 + 4 + 1 = 341 = 11 \times 31$, both primes congruent to 1 modulo 5. $\Phi_3(4) = 4^2 + 4 + 1 = 21 = 3 \times 7$, $3 \equiv 0 \pmod{3}$ and $7 \equiv 1 \pmod{3}$.

Theorem 8

1. Let a and b be positive integers and x an integer. Then $\gcd(x^a - 1, x^b - 1) = |x^{\gcd(a,b)} - 1|$.
2. Let a and b be positive integers. Suppose $\gcd(\Phi_a(x), \Phi_b(x)) > 1$ for some integer x . Then $\frac{a}{b} = p^k$, the integral power of a prime number.

Proof.

1. Let $T = \gcd(x^a - 1, x^b - 1)$ and $t = \gcd(a, b)$. Since $x^t - 1$ divides both $x^a - 1$ and $x^b - 1$. Thus $x^t - 1 | T$. Now as $\gcd(x, T) = 1$. Let $d = \text{ord}_T(x)$. Then $T | x^d - 1$ and since $x^a \equiv x^b \equiv 1 \pmod{T}$, we have $d | a$ and $d | b$, so $d | t$. Thus $x^d - 1 | x^t - 1$, so $T | x^t - 1$. Together we have the result.
2. There are several steps to prove the statement. Suppose p is a common prime divisor of $\Phi_a(x)$ and $\Phi_b(x)$, we want to show $\frac{a}{b}$ is a power of p .
Now let $a = p^\alpha A$ and $b = p^\beta B$, where $\alpha, \beta \geq 0$ and A and B are not divisible by p , we want to show $A = B$.
Clearly $p | \Phi_a(x) | x^a - 1$, thus $p \nmid x$.

(Step 1) We now want to show in fact $p | \Phi_A(x)$. This is trivial if $\alpha = 0$.

Otherwise $\alpha > 0$. By Theorem 6.2, we have

$$0 \equiv \Phi_a(x) = \frac{\Phi_A(x^{p^\alpha})}{\Phi_A(x^{p^{\alpha-1}})} \pmod{p},$$

So $\Phi_A(x^{p^\alpha}) \equiv 0 \pmod{p}$. But $x^{p^\alpha} \equiv x \cdot x^{p^\alpha-1} = x \cdot x^{(p-1)s} \equiv x \pmod{p}$, where s is some integer. Thus $\Phi_A(x) \equiv 0 \pmod{p}$. Likewise $p | \Phi_B(x)$.

(Step 2) Suppose $A > B$. Let $t = \gcd(A, B)$. Then $t < A$. Now $p | \Phi_A(x) | x^A - 1$

and $p \mid \Phi_B(x) \mid x^B - 1$, so $p \mid \gcd(x^A - 1, x^B - 1) = \mid x^t - 1 \mid$. But

$0 \equiv x^t - 1 = \prod_{d \mid t} \Phi_d(x) \pmod{p}$, thus there exists a divisor d of t such that $p \mid \Phi_d(x)$.

But $d \mid t \mid A$, $d < A$ and $p \mid \Phi_A(x)$, so by Theorem 7.1, $p \mid A$, a contradiction.

Likewise $A < B$ is not true. Thus $A = B$.

Theorem 9: If a, n are relatively prime integers, then $\Phi_n(x^a) = \prod_{d \mid a} \Phi_{nd}(x)$.

Proof: By simple counting, both side have the same degree. Thus it suffices to show that every a^{th} root of a primitive n^{th} root of unity is a root of the right hand side.

Indeed a root of the left hand side is of the form ω_{an}^k , with $\gcd(k, n) = 1$ and ω_{an}

a primitive an^{th} root of unity. Let $g = \gcd(k, a)$. Then ω_{an}^k is a primitive

$(\frac{an}{g})^{\text{th}}$ root of unity. If $d = \frac{a}{g}$, then ω_{an}^k is a root of $\Phi_{nd}(x)$. Hence both sides

are equal and we are done.

Applications

Theorem 10: Let d be a positive integer, there exist infinitely many primes p in the arithmetical progression $1, 1+d, 1+2d, \dots, 1+kd, \dots$.

Proof. Suppose not, let $p_1, p_2, \dots, p_r$ be all such primes such that $p_i \equiv 1 \pmod{d}$ for $i = 1, 2, \dots, r$. Define $N = dp_1 p_2 \dots p_r$ so $\mid \Phi_d(N) \mid > 1$. Now let p be a prime such

that $p \mid \Phi_d(N)$. Since $\Phi_d(N) \equiv \Phi_d(0) \equiv 1 \pmod{N}$, $p \nmid N$, $p \nmid d$. Thus by Theorem

7.1, $p \equiv 1 \pmod{d}$, but clearly p is not one of the p_i 's.

Example 5 (IMO2002 Shortlist). Let $p_1, p_2, \dots, p_n$ be distinct primes greater than 3. Then $2^{p_1 p_2 \dots p_n} + 1$ has at least 2^{n-1} divisors.

Proof. It suffices to show that $2^{p_1 p_2 \dots p_n} + 1$ has at least 2^{n-1} pairwise relatively prime divisors and hence at least 2^{n-1} divisors. We know

$$\begin{aligned}
(2^{p_1 p_2 \dots p_n} + 1)(2^{p_1 p_2 \dots p_n} - 1) &= 2^{2 p_1 p_2 \dots p_n} - 1 = \prod_{d|2 p_1 p_2 \dots p_n} \Phi_d(2) \\
&= \left(\prod_{d|p_1 p_2 \dots p_n} \Phi_d(2) \right) \left(\prod_{d|p_1 p_2 \dots p_n} \Phi_{2d}(2) \right) \\
&= (2^{p_1 p_2 \dots p_n} - 1) \left(\prod_{d|p_1 p_2 \dots p_n} \Phi_{2d}(2) \right),
\end{aligned}$$

and hence $2^{p_1 p_2 \dots p_n} + 1 = \prod_{d|p_1 p_2 \dots p_n} \Phi_{2d}(2).$

From Theorem 8.2, if $\Phi_a(2)$ and $\Phi_b(2)$ are not relatively prime, then a/b is a prime power. We just have to find 2^{n-1} divisors of $p_1 p_2 \dots p_n$ so that no two of them differ by exactly one prime number. We just to take those divisors who contain even number of prime divisors, and there are 2^{n-1} of them.

Example 6 (IMO2006 Shortlist). Find all integer solutions of the equation

$$\frac{x^7 - 1}{x - 1} = y^5 - 1.$$

The equation is equivalent to

$$1 + x + x^2 + \dots + x^6 = (y - 1)(1 + y + \dots + y^4).$$

From Theorem 7.2, if p divides $1 + x + \dots + x^6$, then either $p = 7$ or $p \equiv 1 \pmod{7}$. Thus $y - 1 \equiv 0 \pmod{7}$ or $y - 1 \equiv 1 \pmod{7}$; hence $y \equiv 1 \pmod{7}$ or $y \equiv 2 \pmod{7}$. If $y \equiv 1 \pmod{7}$, then $1 + y + \dots + y^4 \equiv 5 \not\equiv 0, 1 \pmod{7}$, a contradiction. If $y \equiv 2 \pmod{7}$, then $1 + y + \dots + y^4 \equiv 31 \equiv 3 \not\equiv 0, 1 \pmod{7}$, another contradiction. Thus the equation has no solution.

Example 7 (USAMO1977). Determine all pairs of integers (m, n) such that

$$1 + x^n + x^{2n} + \dots + x^{mn} \text{ is divisible by } 1 + x + x^2 + \dots + x^m.$$

The expressions are equivalent to $\frac{x^{n(m+1)} - 1}{x^n - 1}$ and $\frac{x^{m+1} - 1}{x - 1}$, respectively. Thus

$$\frac{(x^{n(m+1)} - 1)(x - 1)}{(x^{m+1} - 1)(x^n - 1)}$$

a polynomial. Now the roots of $x^{n(m+1)} - 1$ are distinct, so $x^{m+1} - 1$ and $x^n - 1$ have no common factor other than $(x - 1)$, hence $\gcd(m+1, n) = 1$. The condition is sufficient since $x^{n(m+1)} - 1 = (x^n)^{m+1} - 1 = (x^{m+1})^n - 1$ thus divisible by both $x^n - 1$ and $x^{m+1} - 1$.

Example 8 (IMO1995.6) Let p be an odd prime number. Find the number of subsets

A of the set $\{1, 2, \dots, 2p\}$ such that

- (a) A has exactly p elements, and
(b) the sum of all the elements in A is divisible by p .

Let $\xi = e^{2\pi i/p}$ be a primitive root of unity. The idea is to consider products of p elements within the list $\xi^1, \xi^2, \dots, \xi^{2p}$.

Then

$$\prod_{i=1}^{2p} (x - \xi^i) = (x^p - 1)^2 = x^{2p} - 2x^p + 1.$$

By Viète's formula applied to the coefficient of x^p , get

$$2 = \sum_{1 \leq i_1 \leq \dots \leq i_p \leq 2p} \xi^{i_1 + \dots + i_p} = \sum_{j=0}^{p-1} n_j \xi^j,$$

Where n_j is the number of subsets $\{i_1, i_2, \dots, i_p\}$ of $\{1, 2, \dots, 2p\}$ such that

$$i_1 + i_2 + \dots + i_p = j \pmod{p}.$$

We look for the number n_0 .

But ξ is also a root of the polynomial of degree $p-1$, namely

$$P(x) = n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j.$$

On the other hand, the minimal polynomial of ξ is the cyclotomic polynomial

$$\phi_p(x) = 1 + x + x^2 + \dots + x^{p-1}.$$

It follows that they differ by a constant c ,

$$n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j = c(1 + x + x^2 + \dots + x^{p-1}).$$

By setting $x=1$, we find

$$c = \frac{1}{p} \left(\sum_{j=0}^{p-1} n_j - 2 \right) = \frac{1}{p} \left(\binom{2p}{p} - 2 \right).$$

Using c , we obtain the value of n_0 by comparing the constant terms:

$$n_0 = 2 + \frac{1}{p} \left(\binom{2p}{p} - 2 \right).$$

Theorem 11: Every finite division ring is commutative.

Proof: Let F be the centre of the finite division ring A and let $|F| = q, [A : F] = n$.

Then $|A| = q^n$. We have to show that $n = 1$. Let A^* be the multiplicative group of nonzero elements of A . Then we have the class equation

$$|A^*| = q^n - 1 = \sum_i [A^* : \text{Stab } x_i]$$

where the x_i range over a set of representatives of the *conjugacy* class of A^* (one element from each class). If $x_i \in F$, then $\text{Stab } x_i = A^*$, so we have a contribution of

1 of the above sum coming from such an x_i . The number of $x_i \in F \cap A^*$ is $q - 1$, so altogether we get the contribution $q - 1$ in this way. Now let $x_i \notin F$. Then the subset of elements of A^* is a division subring F_i of A containing F . Hence

$|F_i| = q^{d_i}$ where $d_i = [F_i : F]$ and $d_i < n$ since $x_i \notin F$ and so $F_i \subset A$. It is clear

that $|\text{Stab } x_i| = q^{d_i} - 1$. And so we have the equation re-written as

$$q^n - 1 = (q - 1) + \sum_i \frac{q^n - 1}{q^{d_i} - 1}$$

where every $d_i < n$. Indeed $d_i | n$, since F may be regarded as a (left) vector space

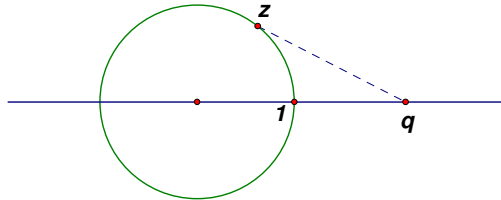
over F_i . Likewise F_i is in the usual way as a vector space over F . Thus $d_i | n$.

Thus then we see $\Phi_n(q)$ divides the left hand side and also every term in the sum.

Hence $\Phi_n(q)$ divides $q - 1$. But note $\Phi_n(q) = \prod (q - z)$, where z runs over the

n^{th} primitive roots of 1. However if $n > 1$, then $|q - z| > q - 1$, contrary to

$\Phi_n(q) | q - 1$.



Exercises

1. p a prime and m a positive integer. If $p \nmid m$, then $\Phi_{pm}(x)\Phi_m(x) = \Phi_m(x^p)$.
2. n odd, ω a primitive n th root of unity if and only if $-\omega$ a primitive $2n$ th root of unity. Hence $\Phi_{2n}(x) = \Phi_n(-x)$.
3. (Wedderburn) Every finite division ring (every nonzero element has an inverse) is a field (commutative).
4. Factorize $5^{1985} - 1$ into product of three divisors such that each is larger than 5^{100} .
4. Note $1985 = 397 \times 5$. Let $x = 5^{397}$. Now $x^5 - 1 = (x-1)(x^4 + x^3 + x^2 + x + 1)$. And

$$\begin{aligned} x^4 + x^3 + x^2 + x + 1 &= x^4 + (6x^3 - 5x^3) + (11x^2 - 10x^2) + (6x - 5x) + 1 \\ &= (x^2 + 3x + 1)^2 - 5x(x+1)^2 \\ &= (x^2 + 3x + 1)^2 - (5^{199}(x+1))^2 \\ &= ((x^2 + 3x + 1)^2 + 5^{199}(x+1))((x^2 + 3x + 1)^2 - 5^{199}(x+1)). \end{aligned}$$

Plug in the value of x to get three factors as desired.

(Remark: Not much to do with the cyclotomic polynomials, and that factorization of polynomials differs from that of integers.)

5. The sequence 10001, 100010001, 1000100010001, ... contains no prime number.
5. Note $10001 = 73 \times 173$, and

$$\begin{aligned} 100010001 &= 10^8 + 10^4 + 1 \\ &= 10^8 + 2 \times 10^4 + 1 - 10^4 \\ &= (10^4 + 1)^2 - (10^2)^2 \\ &= ((10^4 + 1) + (10^2))((10^4 + 1) - (10^2)). \end{aligned}$$

We want to show $10^{4n} + \dots + 10^8 + 10^4 + 1$ are not primes. Clearly if $m+1 \mid n+1$, then $10^{4m} + \dots + 10^8 + 10^4 + 1 \mid 10^{4n} + \dots + 10^8 + 10^4 + 1$, thus it suffices to consider $n+1$ primes. Now

$$10^{4n} + \dots + 10^8 + 10^4 + 1 = \Phi_{n+1}(10^4) = \Phi_{n+1}(10) \cdot \Phi_{2n+2}(10) \cdot \Phi_{4n+4}(10), \text{ and hence not prime.}$$

6. Let n be a positive integer. Show that the number $2^{2^n} + 2^{2^{n-1}} + 1$ can be expressed as the product of no less than n prime factors (not necessarily different).
6. $2^{2^n} + 2^{2^{n-1}} + 1 = \Phi_3(2^{2^{n-1}}) = \prod_{d \mid 2^{n-1}} \Phi_{3d}(2)$. Note that $\Phi_{3d}(2) > 1$ and

$\gcd(\Phi_{3d}(2), \Phi_{3d'}(2)) = 1$ for we know the gcd must be a prime power and as $d \nmid d'$ is a power of 2, then so is the gcd but it is absurd since neither if its expression is divisible by 2 since $\Phi_n(x)$ having a constant term of ± 1 for all n . Thus the claim

is established.

7. Prove that there exist infinitely many positive integers n such that all prime divisors of $n^2 + n + 1$ are not greater than $\sqrt{n}$.

7. Look for $n = k^m$, where $\gcd(m, 3) = 1$. Now $n^2 + n + 1 = \Phi_3(k^m) = \prod_{d|m} \Phi_{3d}(k)$.

Note that $(k+1)^{\varphi(3n)} > \Phi_{3n}(k)$ for all n since k is at most $k+1$ away from each

primitive $3n^{\text{th}}$ roots of unity. Thus we seek to show for some k , there exists an m such that $(k+1)^{\varphi(3m)} < k^{m/2}$ and the result follows because each term in the product

is at most $\sqrt{n}$, implying the result. As $\frac{\varphi(n)}{n}$ can get arbitrarily close to 0, choose

an $n > 1000$ and that $\frac{\varphi(n)}{n} < 0.01$. Then by letting $m = n$, we have

$(k+1)^{\varphi(3m)} < (k+1)^{0.01m}$. Clearly for some k we have $(k+1)^{0.01m} < k^{m/2}$ by making k arbitrarily large, so we are done.

8. Given a positive integer $n > 1$, we have $\Phi_n(1) = 1$ if n is not a prime power and if n is a power of p then we have $\Phi_n(1) = p$.

8. By induction the case is true for $n = 2$. So suppose the case is true for all $k < n$. Now $x^{n-1} + x^{n-2} + \dots + 1 = \prod_{\substack{d|n \\ d \neq 1}} \Phi_d(x)$. Plugging in $x = 1$, get $n = \prod_{\substack{d|n \\ d \neq 1}} \Phi_d(1)$. If n is

a prime power p^s , say, then by the induction hypothesis

$\Phi_p(1) = \Phi_{p^2}(1) = \dots = \Phi_{p^{s-1}}(1) = p$ and we get $p^s = p^{s-1} \Phi_{p^s}(1)$, and hence

$\Phi_{p^s}(1) = p$. If $n = p_1^{s_1} p_2^{s_2} \dots p_r^{s_r}$, $r > 1$, then $\Phi_{p_i^{s_i}}(1) = p_i$, $t_i \leq s_i$, and those products

multiplied to get n and all other products must be reduced to 1, in particular $\Phi_n(1) = 1$.

(Remark: See Sun's article to see examples of how it can be applied, using Galois's Theory).

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